FACET INTO MULITIPLE VIEWS

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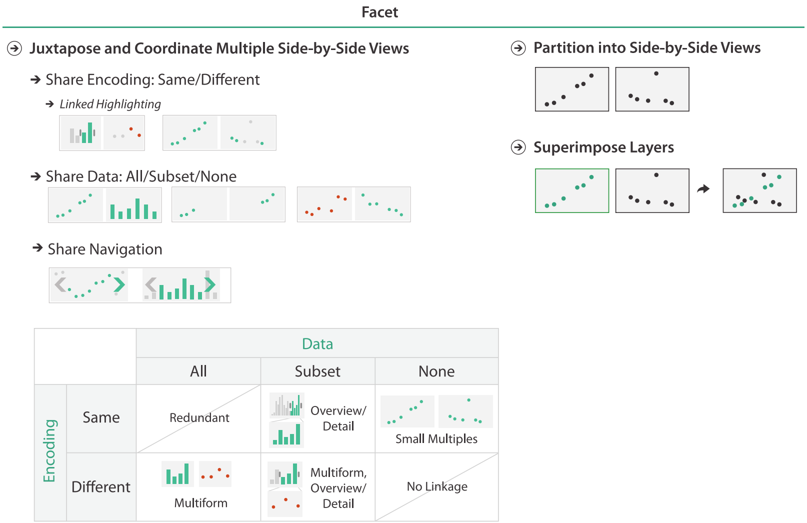
The Big Picture



Outline

- This lecture covers choices about how to facet data across multiple views
- One option for showing views is juxtapose them side by side, leading to many choices of how to coordinate these views with each other.
- The other option is to superimpose the views as layers on top of each other

Why Facet?

Introduction

- The verb **facet** means to split. The design choices pertain to splitting up the display, into either multiple views or layers
- The data needs to be partitioned into the views or layers.
- A major difference between layering and juxtaposing is the strong limits on the number of layers that can be superimposed on each other before the visual clutter becomes overwhelming: two is very feasible and three is possible with care, but more would be difficult. In contrast, the juxtaposing choice can accommodate a much larger number of views, where several is straightforward and up to a few dozen is viable with care.

Juxtapose and Coordinate Views

- Share Encoding: Same/Different
- Share Data: All, Subset, None
- Share Navigation: Synchronize
- Combinations
- Juxtapose Views

Introduction

- Using multiple juxtaposed views involves many choices about how to coordinate between them to create **linked views**
- There are four major design choices for how to establish some sort of linkage between the views.

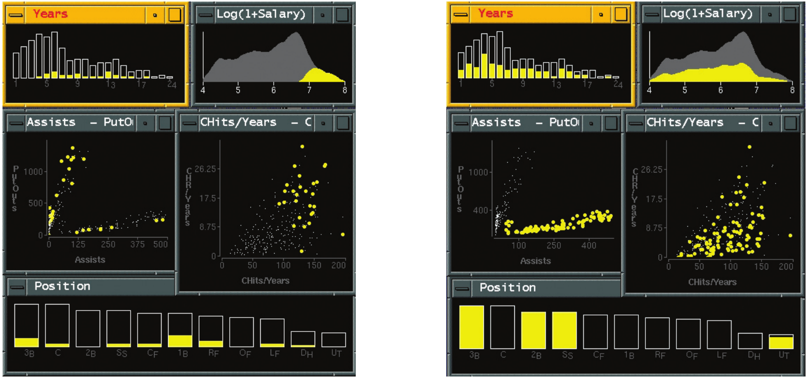
Share Encoding: Same/Different

The most common method for linking views together is to have some form of shared visual encoding where a visual channel is used in the same way across the multiple views.

- The design choice of **shared encoding views** means that all channels are handled the same way for an identical visual encoding
- The design choice of **multiform views** means that some, but not necessarily all, aspects of the visual encoding differ between the two views
- One of the most common forms of linking is **linked highlighting**, where items that are interactively selected in one view are immediately highlighted in all other views using in the same highlight color

Example: Exploratory Data Visualizer (EDV)

- The EDV system features the idiom of linked highlighting between views



(a) (b)

Idiom

System	Exploratory Data Visualizer (EDV)
What: Data	Tables.
How: Encode	Bar charts, scatterplots, and histograms.
How: Facet	Partition: multiform views. Coordinate: linked high-lighting.

Share Data: All, Subset, None

A second design choice is how much data is shared between the two views. There are three alternatives:

- with **shared data**, both views could each show all of the data
- with **overview–detail**, one view could show a subset of what is in the other
- with **small multiples**, the views could show different partitions of the dataset into disjoint pieces.

Example: Bird's-Eye Maps

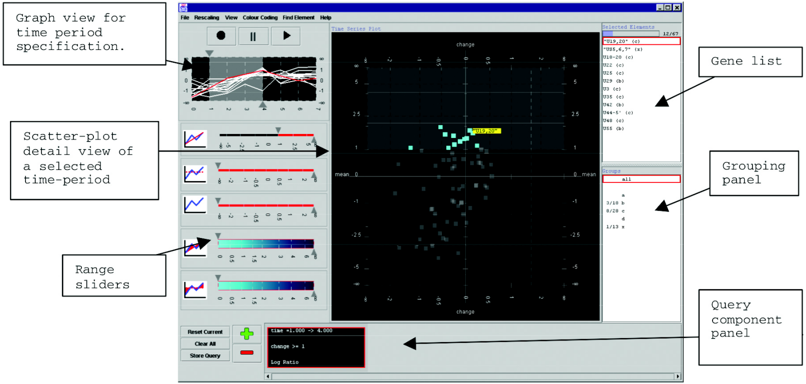
- Overview–detail example with geographic maps, where the views have the same encoding and dataset; they differ in viewpoint and size.

Idiom

Idiom	Bird's-Eye Maps
What: Data	Geographic.
How: Encode	Use given.
How: Facet	Partition into two views with same encoding, overview–detail.
(How: Reduce)	Navigate.

Example: Multiform Overview–Detail Microarrays

- Multiform overview–detail vis tool for microarray exploration features a central scatterplot linked with the graph view in the upper left.

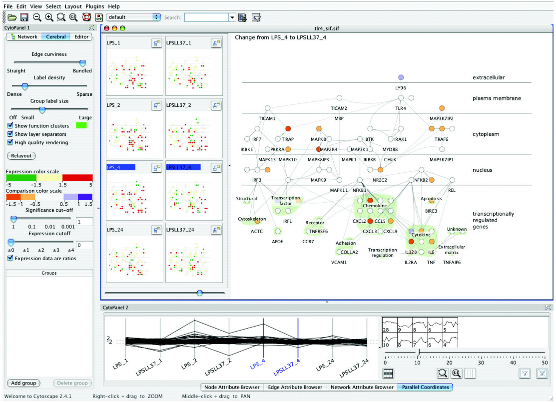


Idiom

System	Multiform Overview–Detail Microarrays
What: Data	Multidimensional table: one categorical key attribute (gene), one ordered key attribute (time), one quantitative value attribute (microarray measurement of gene activity at time).
What: Derived	Three quantitative value attributes: (value change, percentage of max value, fold change).
Why: Tasks	Locate, identify, and compare; distribution, trend, and similarity. Produce.
How: Encode	Line charts, scatterplots, lists.
How: Facet	Partition into multiform views. Coordinate with linked highlighting. Overview+detail filtering of time range. Superimpose line charts.

Example: Cerebral

- Cerebral uses small-multiple views to show the same base graph of gene interactions colored according to microarray measurements made at different times. The coloring in the main view uses the derived attribute of the difference in values between the two chosen views



Idiom

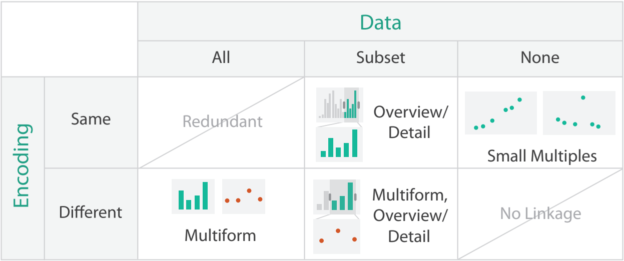
System	Cerebral
What: Data	Multidimensional table: one categorical key attribute (gene), one categorical key attribute (condition), one quantitative value attribute (gene activity at condition). Network: nodes (genes), links (known in teraction between genes), one ordered attribute on nodes: location within cell of interaction.
What: Derived	One quantitative value attribute (difference between measurements for two partitions).
How: Encode	Node–link network using connection marks, vertical spatial position expressing interaction location, containment marks for coregulated gene groups, diverging colormap. Small-multiple network views aligned in matrix. Parallel coordinates.
How: Facet	Partition: small multiple views partitioned on condition, and multiform views. Coordinate: linked high lighting and navigation.

Share Navigation: Synchronize

- Another way to coordinate between views is to **share navigation**. With **linked navigation**, moving the viewpoint in one view is synchronized to movement in the others.

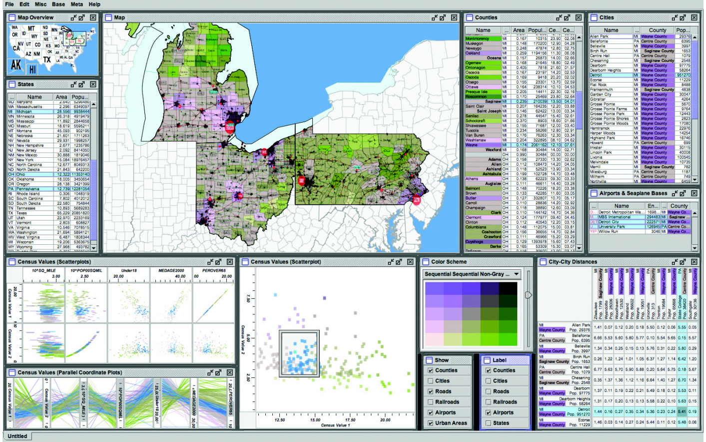
Combinations

- The below figure summarizes the design choices for coordinating views in terms of whether the encoding and data are shared or different and how these choices interact with each other



Example: Improvise

- The Improvise toolkit was used to create this census vis that has many forms of coordination between views. It has many multiform views, some of which use small multiples, and some of which provide additional detail information



Idiom

System	Improvise
What: Data	Geographic and multidimensional table (census data): one key attribute (county), many quantitative attributes (demographic information).
How: Encode	Scatterplot matrix, parallel coordinates, choropleth map with size-coded city points, bird's-eye map overview, scatterplot, reorderable text lists, text matrix. Bivariate sequential–sequential colormap.
How: Facet	Partition: small-multiple, multiform, overview–detail views; linked highlighting.

Juxtapose Views

- Two additional design choices that pertain to view juxtaposition do not directly involve view coordination: when to show each view and how to arrange them.
  - A view temporarily pops up in response to a user action.
  - Arrange the views themselves automatically

Partition into Views

- Regions, Glyphs, and Views
- List Alignments
- Matrix Alignments
- Recursive Subdivision

Introduction

The design choice of how to **partition** a multiattribute dataset into meaningful groups has major implications for what kind of patterns are visible to the user

- The primary design choice within partitioning is how to divide the data up between the views, given a hierarchy of attributes
- One design choice is how many splits to carry out
- Another design choice within partitioning is the order in which attributes are used to split things up
- Final design choice is how many views to use
- A partitioning attribute is typically a categorical variable that has only a limited number of unique values; that is, **levels**

Regions, Glyphs, and Views

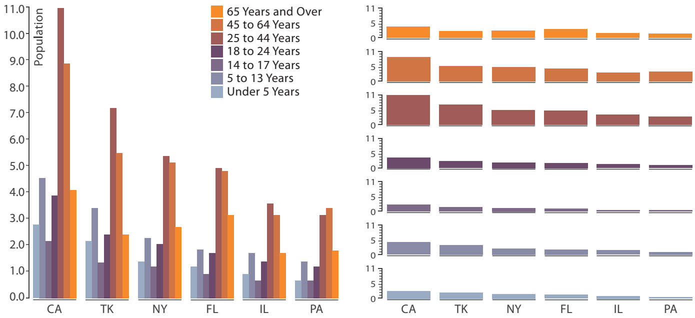
- Partitioning is an action on a dataset that separates data into groups
- To connect partioning to visual encoding choices, the crucial idea is that a partitioned group can be placed within a **region** of space
- These regions then need to be ordered, and often aligned, to resolve the other spatial arrangement choices
- Encoding could be just a single **mark**, a single geometric primitive.
- Encoding could be a more complex **glyph**: an object with internal structure that arises from multiple marks.
- A view is a contiguous region in which visually encoded data is shown on the display

Example

Partitioning and bar charts

(a) Single bar chart with grouped bars: separated by state key into regions, using seven-mark glyphs within each region.

(b) Seven aligned small-multiple bar chart views: separated by group key into vertically aligned list of regions, with a full bar chart in each region



List Alignments

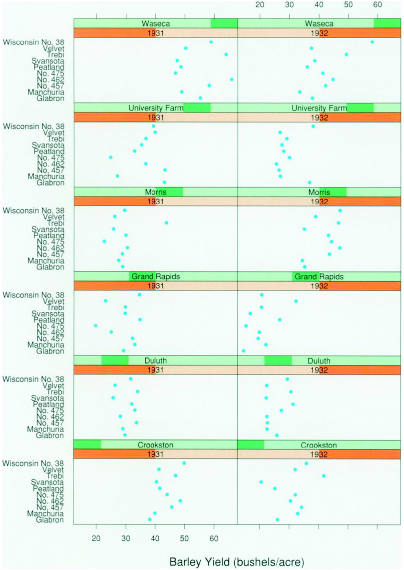
- The grouped bar chart facilitates comparison between the attributes, whereas the small multiple bar charts facilitate comparison within a single attribute
- From a partitioning point of view, both idioms use two levels of partitioning: at the high level by a first key, and then at a lower level by a second key

Example: Trellis

- The Trellis system is a more complex example. This system features partitioning a multiattribute dataset into multiple views and ordering them within a 2D matrix alignment as the main avenue for exploration

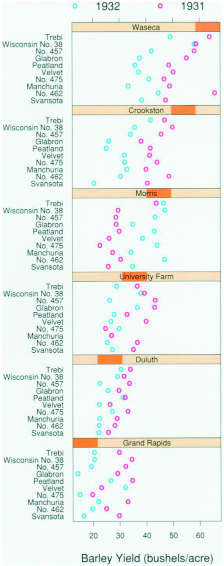
Example: Trellis (cont.)

- Trellis facets the data into a matrix of dot chart views, allowing the user control of partitioning and orderering



Example: Trellis (cont.)

- A second Trellis plot combines the years into a single plot with year encoded by color, showing strong evidence of an anomaly in the data.



Idiom

System	Trellis
What: Data	Multidimensional table: three categorical key attributes, one quantitative value attribute.
What: Derived	Medians for each partition.
How: Encode	Dot charts aligned in 2D matrix.
How: Facet	Partitioned by any combination of keys into regions.

Recursive Subdivision

- Partitioning can be used in an exploratory way, where the user can reconfigure the display to see different choices of partioning and encoding variables
- A dataset of over one million property transactions in the London area.
    - The categorical attribute of *residence type* has four levels: flats *Flat*, attached terrace houses *Ter*, semidetached houses *Semi*, and fully detached houses *Det*.
    - The *price* attribute is quantitative.
    - The *time of sale* attribute is provided as a year and a month, for an ordered attribute with hierarchical internal structure.
    - The *neighborhood* attribute, with 33 levels, is an interesting case that can be considered either as categorical or as spatial.

Recursive Subdivision (cont.)

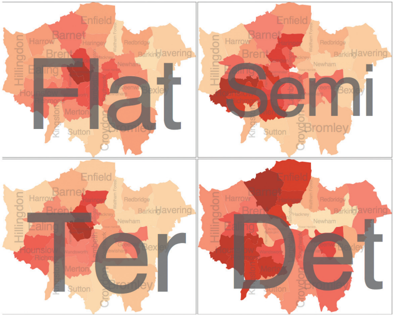
- The HiVE system supports exploration through different partitioning choices.
  - **(a)** Recursive matrix alignment where the first split is by the house type attribute, and the second by neighborhood. The lowest levels show time with years as rows and months as columns.
  - **(b)** Switching the order of the first and second splits shows radically different patterns.

(a) (b)

Recursive Subdivision (cont.)

- HiVE with different arrangements.
  - **(a)** Sizing regions according to sale counts yields a treemap.
  - **(b)** Arranging the second-level regions as choropleth maps.

(a)



(b)

Superimpose Layers

- Visually Distinguishable Layers
- Static Layers
- Dynamic Layers

Introduction

- The **superimpose** family of design choices pertains to combining multiple layers together by stacking them directly on top of each other in a single composite view
- A visual **layer** is simply a set of objects spread out over a region
- The design choices for how to superimpose views include: How many layers are used? How are the layers visually distinguished from each other? Is there a small static set of layers?

Visually Distinguishable Layers

- One good way to make distinguishable layers is to ensure that each layer uses a different and nonoverlapping range of the visual channels active in the encoding
- A common choice is to create two visual layers, a **foreground** versus a **background**
- Layering many views is only feasible if each layer contains very little, such as a single line

Static Layers

- The design choice of static layers is that all of the layers are displayed simultaneously; the user can choose which to focus on with the selective direction of visual attention. Mapmakers usually design maps in exactly this way.

Example: Cartographic Layering

- Static visual layering in maps. (a) The map layers are created by differences in the hue, saturation, luminance, and size channels on both area and line marks. (b) The grayscale view shows that each layer uses a different range in the luminance channel, providing luminance contrast

(a)

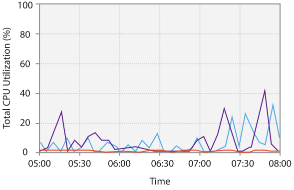
(b)

Idiom

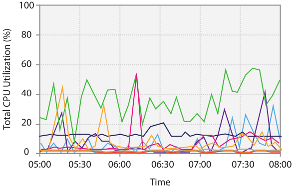
Idiom	Cartographic Layering
What: Data	Geographic
How: Encode	Area marks for regions (water, parks, other land), line marks for roads, categorical colormap.
How: Facet	Superimpose: static layers distinguished with color saturation, color luminance, and size channels.

Example: Superimposed Line Charts

- Multiple line charts can be superimposed within the same global frame. (a) A small number of items is easily readable. (b) Up to a few dozen lines can still be understood. (c) This technique does not scale to hundreds of items.



(a)



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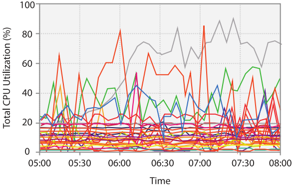


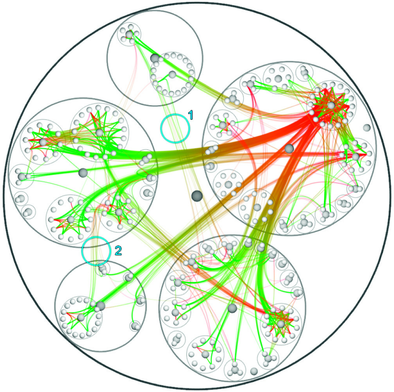
(c)

Idiom

Idiom	Superimposed Line Charts
What: Data	Multidimensional table: one ordered key attribute (time), one categorical key attribute (machine), one quantitative value attribute (CPU utilization).
How: Encode	Line charts, colored by machine attribute.
How: Facet	Superimpose: static layers, distinguished with color.
Scale	Ordered key attribute: hundreds. Categorical key attribute: one dozen.

Example: Hierarchical Edge Bundles

- The hierarchical edge bundles idiom shows a compound network in three layers: the tree structure in back with containment circle marks, the red–green graph edges with connection marks in a middle layer, and the graph nodes in a front layer

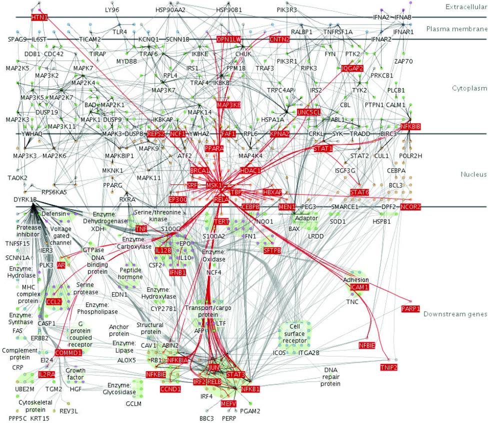


Idiom

Idiom	Hierarchical Edge Bundles
What: Data	Compound graph: network, hierarchy whose leaves are nodes in network.
How: Encode	Back layer shows hierarchy with containment marks colored gray, middle layer shows network links colored red–green, front layer shows nodes colored gray.
How: Facet	Superimpose static layers, distinguished with color.

Dynamic Layers

- With **dynamic layers**, a layer with different salience than the rest of the view is constructed interactively, typically in response to user selection.
- The number of possible layers can be huge



Cerebral dynamically creates a foreground visual layer of all nodes one topological hop away in the network from the node underneath the cursor.

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